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A Machine Learning-Based Groundwater Level Predictor for Sustainable Water Management For Karnataka

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Abstract: This project proposal reports the creation of a machine learning-based 'Groundwater Level Predictor' for Karnataka with emphasis on sustainable water management. The project aims to ease the severe stress on groundwater resources for rising demand and climate change by offering predictive tools for proactive management. Our aim is to develop a strong model that precisely predicts the groundwater level. This includes building an integrated spatio-temporal dataset from varied sources, applying and testing high-level machine learning models (mainly Random Forest), validating their performance strenuously, and deriving actionable insights into the major drivers of groundwater level changes. This project produces a validated predictive model, a complete dataset, and a technical report in detail, all intended to assist governmental agencies in data-driven water allocation, drought preparedness, and policy evaluation for securing Karnataka's water future. With mounting water demand and climate change effects, this effort seeks to offer administrative stewardship tools. The main aim is to develop a credible model predicting monthly groundwater levels at the taluk level, using a spatio-temporal dataset sourced from various sources. The project will deploy and benchmark state-of-the-art machine learning models, Random Forest being a top contender, under strict performance validation. In addition to prediction, it will determine and measure important drivers of changes in groundwater level to produce actionable information. The deliverables will include a validated prediction model, a full dataset, and a technical report with results. These efforts will start to support government agencies using data to inform decisions around water allocations, drought preparedness, and policies, to help ensure the water future and sustainability of Karnataka.

Index Terms -Machine Learning, Groundwater Level Prediction, Random Forest, Sustainable Water Management, Karnataka, Spatio-Temporal Dataset, Streamlit, Time Series Cross-Validation, Water Allocation, Drought Preparedness, Policy Assessment.

I. INTRODUCTION

Groundwater is a significant natural resource utilized for agriculture, industry, and domestic consumption. In India, it contributes to over 60% of irrigation and approximately 80% of drinking water. Rapid depletion due to over-extraction, urbanization, and climate change threatens the available quantity and quality of groundwater. Although organizations like the Central Ground Water Board (CGWB) monitor water levels across nearly 26,000 wells and are deploying Digital Water Level Recorders (DWLRs) systems, is primarily collect and transmit data without predictive capability. Conventional models such as ARIMA are unable to capture the highly non-linear, multi-factor interactions of aquifer systems.

To address this gap, we developed a software application called the “Ground Water Level Predictor,” which integrates meteorological, hydrological, geospatial, and demographic attributes into a single platform. The application uses a Random Forest regression model implemented in Python and deployed through the Streamlit framework to predict groundwater levels in both space and time. It provides an interactive dashboard with maps, time-series graphs, and soil–water depth profiles to help stakeholders understand and plan groundwater use.
They maintain a vast network of nearly 26,000 observation wells and are progressively deploying Digital Water Level Recorders (DWLRs) to enhance data collection. While these systems effectively gather and transmit data, they predominantly lack advanced predictive capabilities. Conventional hydrological models, such as Autoregressive Integrated Moving Average (ARIMA), have proven inadequate in capturing the highly nonlinear and multi-factor interactions inherent within complex aquifer systems, thus limiting their effectiveness in forecasting future groundwater trends.
Recognizing this critical void, a novel software application, aptly named “Ground Water Level Predictor,” has been developed.
This innovative platform addresses the existing limitations by integrating a complete array of data points, including meteorological, hydrological, geospatial, and demographic attributes, into a unified and accessible interface. The application leverages a robust

Random Forest regression model, implemented using Python. This sophisticated model is deployed through the Streamlit framework, enabling it to accurately predict groundwater levels across both spatial and temporal dimensions.

The "Ground Water Level Predictor" offers an interactive dashboard designed to empower stakeholders with actionable insights. This dashboard features dynamic maps that visualize groundwater distribution, time-series graphs that show historical and projected water level fluctuations with detailed soil–water depth profiles to inform about understanding of groundwater processes to enhance decisions and strategic planning for sustainable groundwater uses and management.

II. EXISTING METHODS

A. Current Groundwater Monitoring Practices

Present Monitoring of Groundwater Governmental entities, including the Central Ground Water Board (CGWB) and the Karnataka Ground Water Authority (KGWA), have a wide-ranging network of observation wells across the state. These networks allow for documentation of long-term aquifer condition but provide no predictive capabilities. Decisions are generally made by extrapolating from trends or through some other subjective means, which is not conducive to water management in a proactive manner.

B. Traditional Statistical Models

The initial endeavors for groundwater level predictions relied on time-series models, including ARIMA, linear regression, and moving averages. These models rely on linearity and stationarity assumptions, which are unreasonable, considering the non-linear and multi-factorial character of aquifer systems subject to rainfall variability, soil heterogeneity, and the pressure associated with human populations. Moreover, they are only capable of prediction within a heterogeneous hydrogeological context, like Karnataka.

C. Machine Learning and Deep Learning Approaches

In the last few years, machine learning has been used to address these problems. Tree-based ensemble models, such as Random Forest and Gradient Boosting, can manage mixed numerical and categorical inputs (rainfall, soil type, population, spatial coordinates) and related models consistently outperform classical statistical models. Deep learning approaches, such as Long Short-Term Memory (LSTM) networks, as well as hybrid CNN-LSTM architectures, have been utilized to capture temporal dependencies and spatial attributes in groundwater datasets.

D. Gap in User-Friendly Deployment

Even though they hold promise, there are still many machine learning and deep learning models for groundwater prediction that are limited to research prototypes and have not yet been deployed in a format that could directly assist with planning and policy in a public-facing interactive manner.

E. Need for an Integrated Solution

Currently, Karnataka lacks a cohesive system that includes excellent groundwater datasets, cutting-edge machine learning models, and an interactive dashboard. This project is motivated by this gap, using an interpretable Random Forest regression model, which is implemented using a Streamlit-supported web application. This increases predictive accuracy while also promoting forecasting transparency for planners, researchers, and local communities.

III. PROPOSED METHODOLOGY

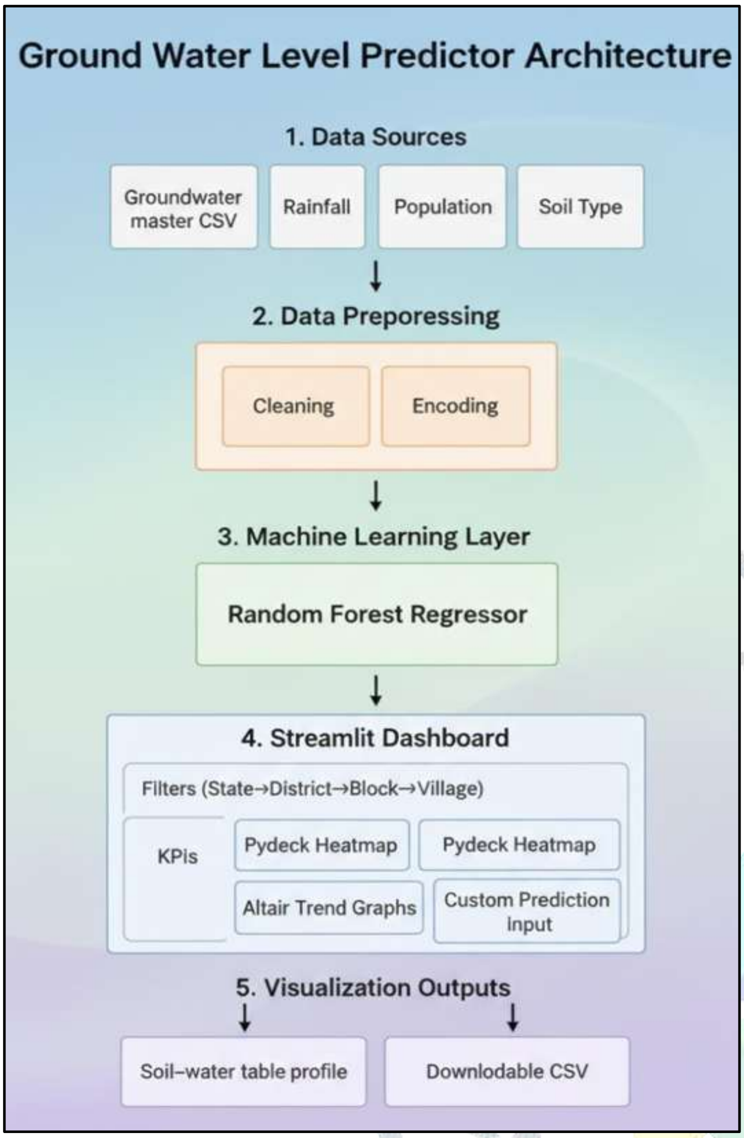

Fig. 1 System Architecture

A. Data Collection

This study employs a consolidated dataset assembled from official sources such as the Central Ground Water Board (CGWB) and the Karnataka Ground Water Authority (KGWA). This data has spatial coordinates (latitude, longitude), observed groundwater levels (WL in meters below ground level), annual rainfall, block population, and predominant soil type for each monitoring location.

B. Data Preprocessing

Before modeling, the raw data is cleaned in a formal cleaning process in Python (pandas):

1. Renaming and standardizing field names
 – (LATITUDE → lat, LONGITUDE → lon).
2. Converting key fields (lat, lon, WL, rainfall, population) to numeric values.
3. Parsing and validating dates.
4. Dropping rows with missing critical values
5. Encoding the categorical variable Predominant_Soil_Type as one-hot encoding for machine learning input.

In this way, the model has consistent, analysis-ready data as inputs.

C. Model Training – Random Forest Regression

The prediction engine of the system is a Random Forest Regressor implemented with scikit-learn. The feature vector consists of rainfall, block population, predominant soil type (encoded), latitude, and longitude. The target variable is the observed groundwater level (WL in mbgl).

- Algorithm: RandomForestRegressor with 100 estimators, random_state=42.
- Advantages: Handles mixed numerical and categorical data, captures non-linear relationships, and is robust to outliers.
- Caching: The trained model is cached with @st.cache_resource to speed up app performance.

D. Dashboard Development – Streamlit Interface:

We built an interactive dashboard using Streamlit to share the outputs of the model in the following way:

I. Hierarchical Filters -Users can drill down from State → District → Block → Village to access data for each administrative unit.

II. Maps - Pydeck heatmaps (for groundwater variation) and scatterplots (showing observation locations) depict locations.

III. Counts of Water Level, Rainfall, Soil Type, and Population - displays matched to scales appear in real time.

Prediction - Users can provide soil type, rainfall, population, latitude, and longitude to estimate a possible water level.

E. Soil and Water Table Visualization:

A custom HTML component renders a color-coded soil profile with a dynamic water table indicator. This helps non-technical users intuitively understand the predicted groundwater depth and underlying soil layers.

F. Validation and Performance Metrics:

To assess prediction reliability, the model was evaluated using common regression indicators — RMSE, MAE, and the coefficient of determination (R^2). These metrics quantify the predictive accuracy and support model tuning.

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G. Deployment and Scalability:

The system is operated entirely as a web application so that planners, researchers, and local communities can access the system without having to download or utilize specialized software. The lightweight architecture of Streamlit allows for easy updates when new data becomes available, and future modeling enhancements such as LSTM or CNN-LSTM for longer-term forecasting.

IV. OBJECTIVES

The main goal of this project is to design and implement a rigorous data-driven system that can predict and visualize groundwater levels for Karnataka using machine learning. The specific goals are as follows:

A. *Data Integration:*

To compile, clean, and integrate groundwater, rainfall, soil type, population, and geospatial data from credible sources into a complete analysis-ready dataset.

B. *Model Development:*

To implement and train a Random Forest regression model that can reliably predict groundwater level (in meters below ground level), rainfall, soil type, population, and spatial coordinates as inputs.

C. *Interactive Dashboard:*

To build a user-friendly Streamlit dashboard that allows end users to filter data by State, District, Block, and Village while viewing actual vs predicted groundwater level.

Fig. 2 Filter Interaction

D. *Visualization and Interpretation:*
To create user-friendly visualizations, Pydeck heatmaps, Altair time-series charts, and a custom soil–water table profile, that result in easily interpretable predictions and underlying data.

E. *Performance Validation:*
To assess the model using standard regression metrics (RMSE, MAE, R_2) and to verify that the predictions are appropriate for making policy and planning decisions.

F. *Scalability and Future Extension:*
To offer a design that allows for the updating of the model with new data, and to extend the model to use more complex models, including LSTM or CNN-LSTM, for long-term forecasting.

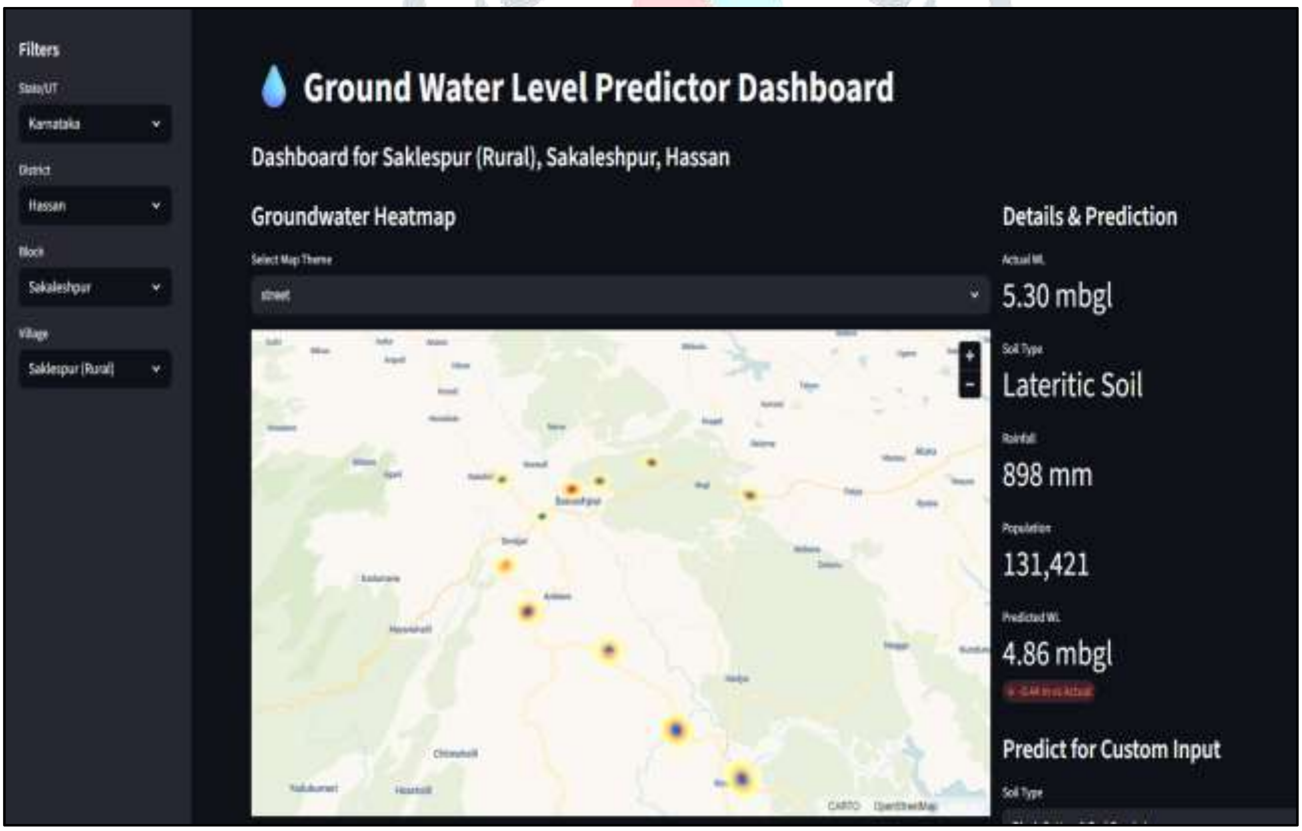

Fig.3 Web Dashboard for Visualization

V. SYSTEM DESIGN AND IMPLEMENTATION

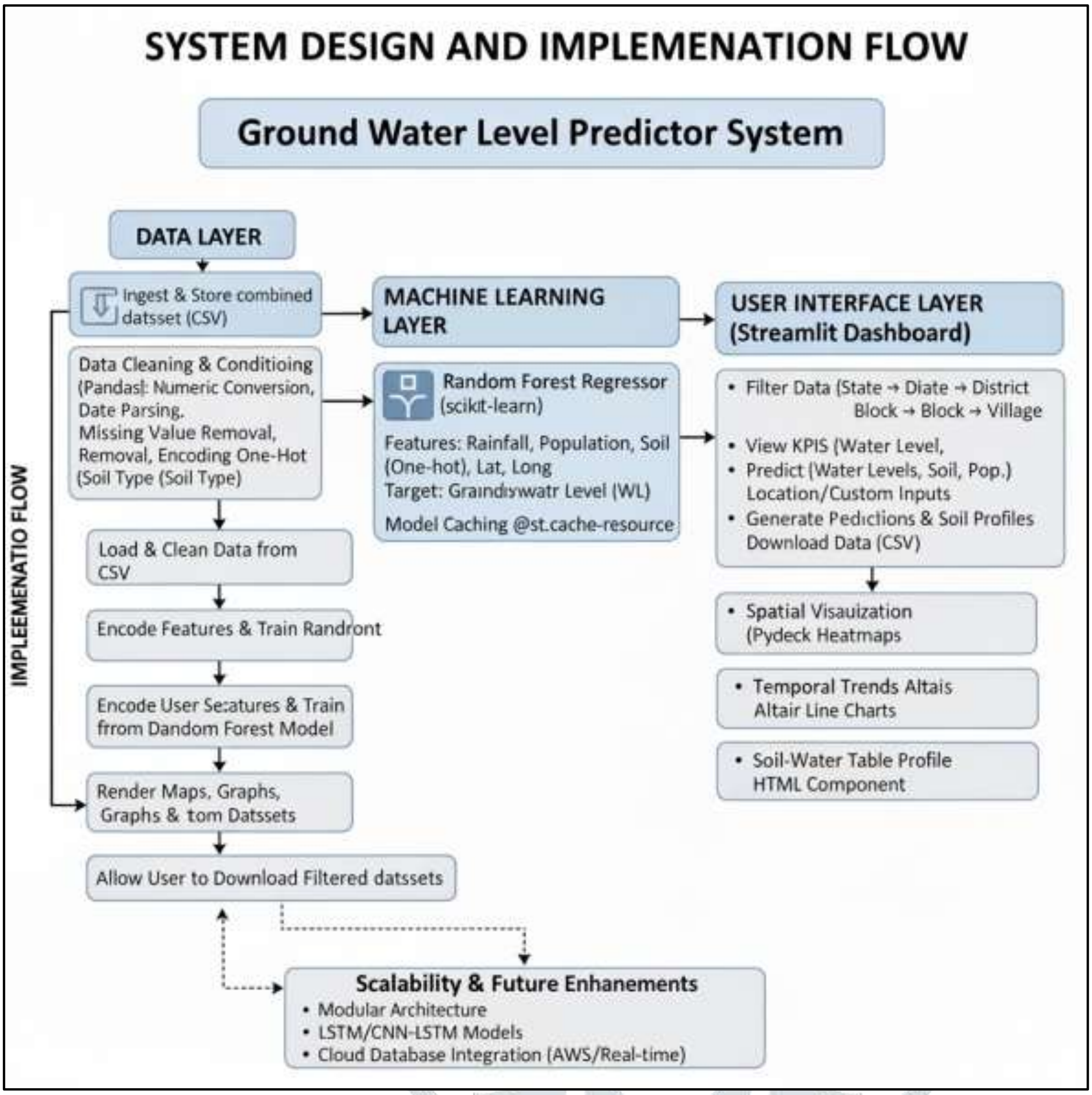

Fig.4 System Design

A. *System Overview*

The Groundwater Level Predictor system functions as a modular pipeline, converting the raw data of groundwater and environmental factors to predictions and to visualisations that can be acted upon, made up of 3 layers: The Data Layer, the Machine Learning Layer, and the User Interface Layer.

B. *Data Layer*

The data layer ingests and stores the combined dataset (groundwater_master_data.csv) for latitude, longitude, observed groundwater levels, yearly rainfall, block population, and the predominant soil type. The data was subsequently cleaned and conditioned for use in the Python (pandas) environment. The data conditioning included Numeric conversion, parsing dates, removal of missing values, and one-hot encoding of soil types.

C. *Machine Learning Layer*

The predictive engine is a Random Forest Regressor using scikit-learn.

- **Features:** Rainfall, Block Population, Predominant Soil Type (One-hot encoding), Latitude, Longitude.
- **Target:** Groundwater level (WL in metres below ground level).
- **Advantages:** Captures non-linear relationships and interactions between features, accommodates mixed feature types, and offers interpretable metrics of feature importance. The model is cached in memory using `@st. cache_resource` to avoid retraining on every user interaction.

D. User Interface Layer (Streamlit Dashboard)

The front end of the system is an interactive Streamlit dashboard, which allows stakeholders to:

1. **Filter Data:** Drill down from State → District → Block → Village.

2. **View KPIs:** Display actual water level, rainfall, soil type, and population metrics for the selected location.

3. **Predict Water Levels:** Generate predicted groundwater levels for the selected village or for custom inputs (soil type, rainfall, population, latitude, longitude).

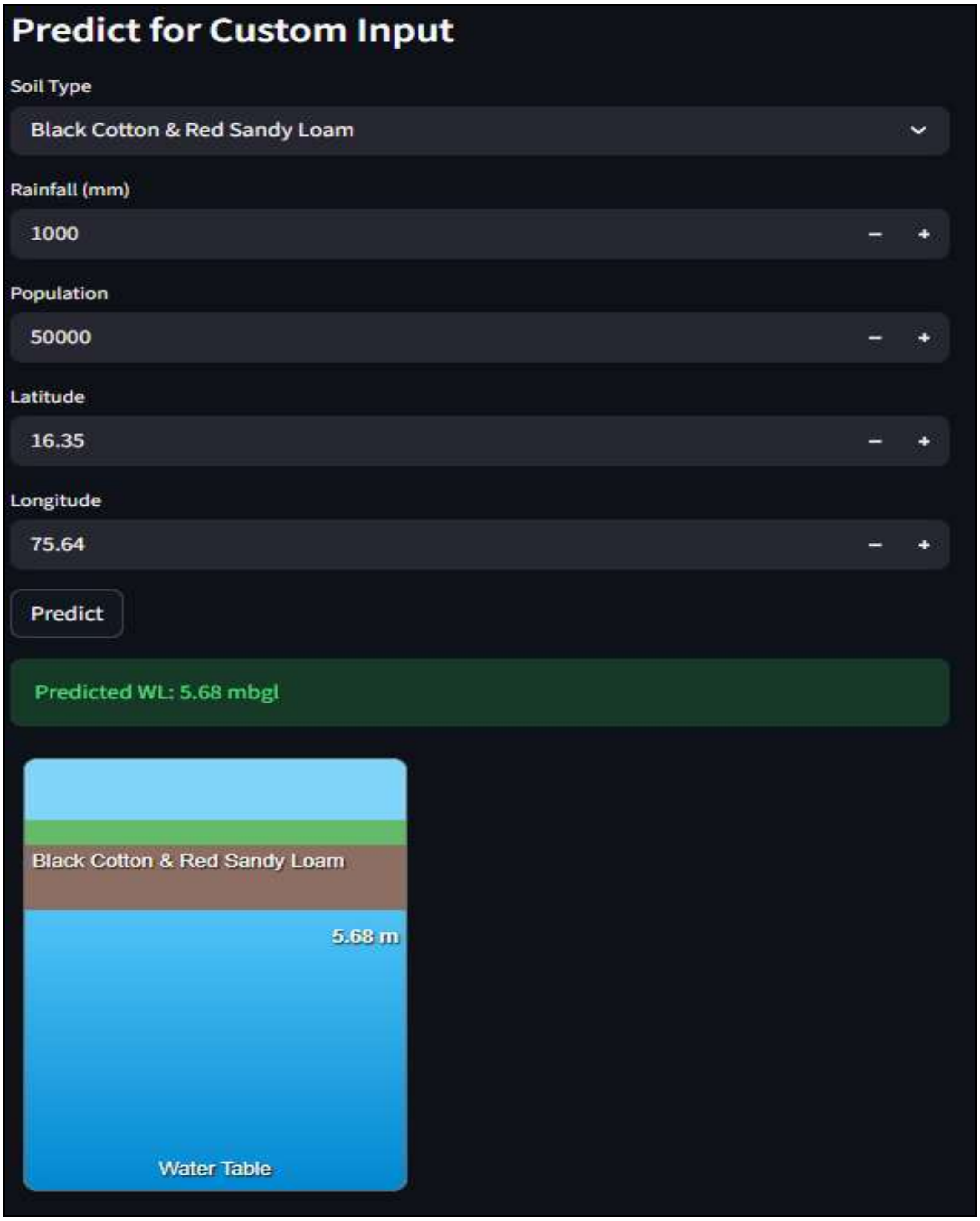

Fig. 5 Water Level Prediction

4. **Download Data:** Export filtered data as CSV for offline analysis.

E. Visualization Components

Two main visualization modules enhance interpretability:

● **Spatial Visualization:** Pydeck heatmaps and scatter plots show spatial variation in groundwater levels and highlight observation points.

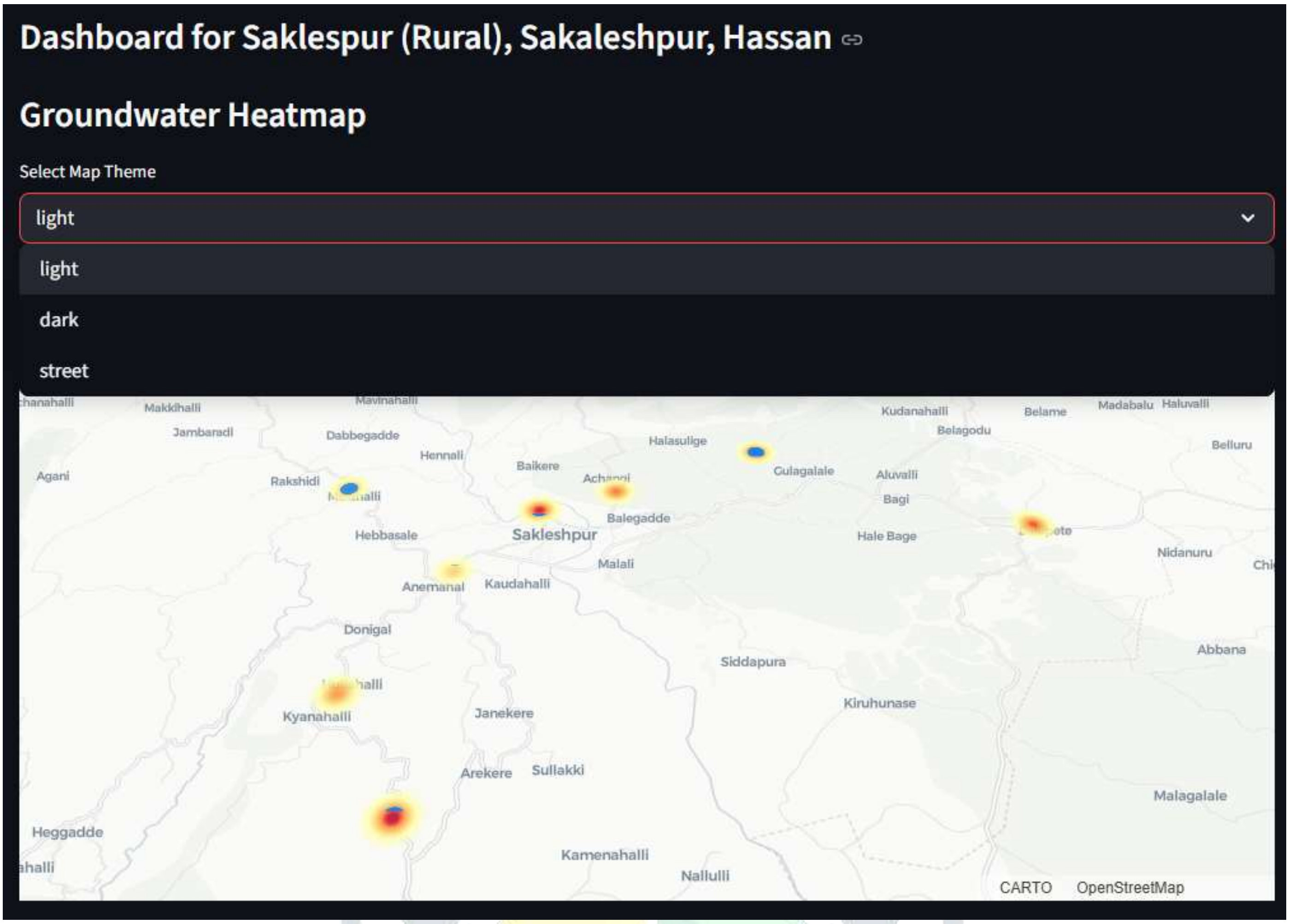

Fig. 6 HeatMap Visualization

● **Temporal Trends:** Altair line charts show monthly or yearly groundwater trends at the district level.

● **Soil–Water Table Profile:** A custom HTML component renders colour-coded soil layers with a dynamic water-table indicator based on predicted depth.

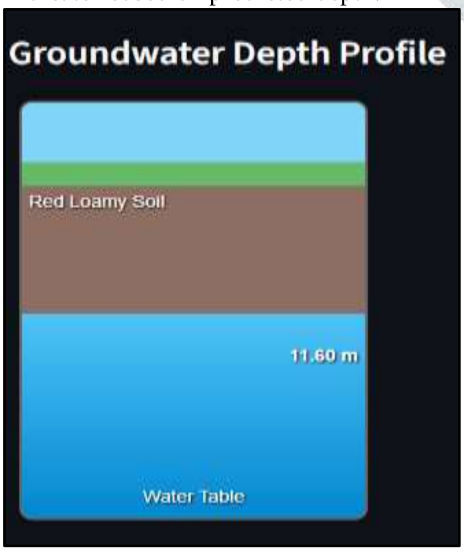

Fig. 7 Groundwater Depth Profile

F. Implementation Flow

The system workflow proceeds as follows:

1. Load and clean data from the CSV file.

2. Encode features and train the Random Forest model.
3. Accept user selections and/or custom inputs from the dashboard.
4. Generate predictions and KPIs.
5. Render maps, graphs, and soil profiles.
6. Allow the user to download filtered datasets.

G. Scalability and Future Enhancements

The use of modular architecture permits the system to be updated with new data and customized using leading-edge models like LSTM or CNN-LSTM for updated long-term forecasting, more geospatial layers built in, and one of the capabilities is also integrated with cloud databases like AWS or similar for real-time operation.

VI. CONCLUSION

Demonstrating water resource sustainability through efficient groundwater management, the Ground Water Level Predictor described in this paper shows how machine learning can be applied with tools such as interactive dashboards to understand and predict groundwater levels at a fine scale. The synthetic use of groundwater observations with rainfall, soil type, population, and geospatial coordinates shows how a Random Forest regression model can provide trustworthy predictions of groundwater levels at a fine scale.

A Stream lit-based dashboard provides all stakeholders with a simple and yet powerful interface for reviewing historical data, generating customized forecasts, and especially visualizing soil–water profiles and spatial patterns. This level of accessibility connects the gap between advanced analytics and decision-making processes in the field to provide planners, researchers, and local communities with data-driven water management solutions.

The current project demonstrates that tree-based ensemble methods (Random Forest) provide acceptable benchmarks for groundwater prediction methods in Karnataka, outperforming traditional statistical models with the ability to capture complex non-linear relationships and municipality-level forecasts. The modular design would also allow additional data sources to be added or an upgrade to more sophisticated deep learning models (e.g., LSTM or CNN-LSTM) for long-term forecasting.

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